

Overview: The Minesweeper program is split into separate files for the user interface, game logic, board data, cell data, and configuration. The user interacts with the GUI or terminal, and those inputs are sent to the Board class to update the game.

System Architecture Diagram



After the board changes, the GUI or terminal redraws the updated board for the user.

Main Components

- **User Interface / Input Handler (gui.py and game.py):** Gets mouse clicks or keyboard commands from the player, shows the board, remaining flags, and win/loss status.
- **Board / Game Logic (board.py):** Creates the board, places mines, reveals cells, handles flags, counts nearby mines, and checks for a win or loss.
- **Cell (cell.py):** Stores the state of one square: mine, revealed, flagged, and adjacent mine count.
- **Configuration (config.py):** Stores shared settings such as the 10×10 board size, A–J column labels, 10–20 mine limit, and first-click safety.

Data Flow

1. The player clicks a cell in the GUI or enters a command in the terminal.
2. gui.py or game.py sends the selected row and column to the Board class.
3. board.py updates the correct Cell objects by revealing, flagging, or placing mines.
4. The interface reads the updated board and redraws the game for the player.

Key Data Structures

- **10×10 2D list:** Board.grid stores 100 Cell objects.
- **Cell object:** Stores is_mine, is_revealed, is_flagged, and adjacent_mines.
- **Game state:** The program tracks mine count, flags remaining, whether the game is over, and whether the board is cleared.